

CHAPTER-I-INTRODUCTION TO AI

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OBJECTIVE

- Understand The History of AI
- Understand AI Definition
- Understand Humanly and Rationally of Thinking and Acting

ARTIFICIAL INTELLIGENCE

1. We are human, the intelligence is our dominance
2. For thousands of years, we have tried to understand “*how we think?*”
3. The field of artificial intelligence (AI) attempts not just to understand but also to build intelligent entities
4. AI is one of the newest fields in science and engineering, started after World War II, coined in 1956 , in Dartmouth Conference by John McCarthy

ARTIFICIAL INTELLIGENCE

5. AI encompasses a huge variety of subfields, from general (learning and perception)
 - such as playing chess, proving mathematical theorems, writing poetry, autonomous car, and diagnosing diseases
6. AI is relevant to any intellectual task; it is a universal field.

HISTORY OF AI

1. The gestation of artificial intelligence (1943–1955)
2. The birth of artificial intelligence (1956)
3. Early enthusiasm, great expectations (1952–1969)
4. A dose of reality (1966–1973)
5. Knowledge-based systems: The key to power? (1969–1979)
6. AI becomes an industry (1980–present)
7. The return of neural networks (1986–present)
8. AI adopts the scientific method (1987–present)
9. The emergence of intelligent agents (1995–present)
10. The availability of very large data sets (2001–present)

I. THE GESTATION OF ARTIFICIAL INTELLIGENCE (1943–1955):

- During this period, early foundations of AI were laid.
- Alan Turing developed the concept of a "universal machine" capable of performing any computation.
- Warren McCulloch and Walter Pitts published a paper on neural networks as a model of computation.
- The groundwork for AI and digital computers was established.

2.THE BIRTH OF ARTIFICIAL INTELLIGENCE (1956):

- The term "artificial intelligence" was coined at the Dartmouth Conference, marking the formal beginning of AI research.
- Researchers like **John McCarthy**, Marvin Minsky, Nathaniel Rochester, and Claude Shannon played key roles in organizing the conference.
- The conference discussed the possibility of creating machines with human-like intelligence.

3. EARLY ENTHUSIASM, GREAT EXPECTATIONS (1952–1969):

- This period was marked by optimism and ambitious goals in AI research.
- Early AI programs like the Logic Theorist and General Problem Solver were developed.
- Researchers aimed to replicate human problem-solving and reasoning abilities in computers.

4.A DOSE OF REALITY (1966–1973):

- The field experienced a setback and reduced funding, referred to as the "**AI winter**."
- Initial expectations of **AI surpassing human intelligence** were tempered as challenges became more apparent.
- Research shifted towards more **practical applications**, such as **expert systems**, **decision tree**, **stock market prediction**

5. KNOWLEDGE-BASED SYSTEMS: THE KEY TO POWER? (1969–1979):

- AI research saw a shift towards **knowledge-based systems**, focusing on representing and using human knowledge.
- Expert systems like **MYCIN**, used for medical diagnosis, gained attention.
 - MYCIN was written in LISP to diagnose a symptom by matching in database
- The development of **rule-based systems** allowed AI to make complex decisions based on human expertise.

6. AI BECOMES AN INDUSTRY (1980–PRESENT):

- The 1980s marked a resurgence in AI research, with applications in various industries, including finance, healthcare, and manufacturing.
- AI systems like **rule-based expert systems** were integrated into **practical applications**.
- AI adoption **increased** in areas like **natural language processing** and **robotics**.

7. THE RETURN OF NEURAL NETWORKS (1986–PRESENT):

- **Neural networks**, which had been less popular since the 1960s, experienced a resurgence in the 1980s, thanks to developments in **backpropagation** and the availability of data.
- **Deep learning** and **neural networks** became **central to AI**, leading to significant advancements in areas like image and speech recognition.

8. AI ADOPTS THE SCIENTIFIC METHOD (1987–PRESENT):

- The field adopted more **rigorous scientific methodologies** and evaluation criteria.
- Research in AI **ethics**, fairness, and bias **gained prominence**.
- AI applications in areas like **healthcare** and **autonomous vehicles** became more **reliable** and **safer**.

9. THE EMERGENCE OF INTELLIGENT AGENTS (1995–PRESENT):

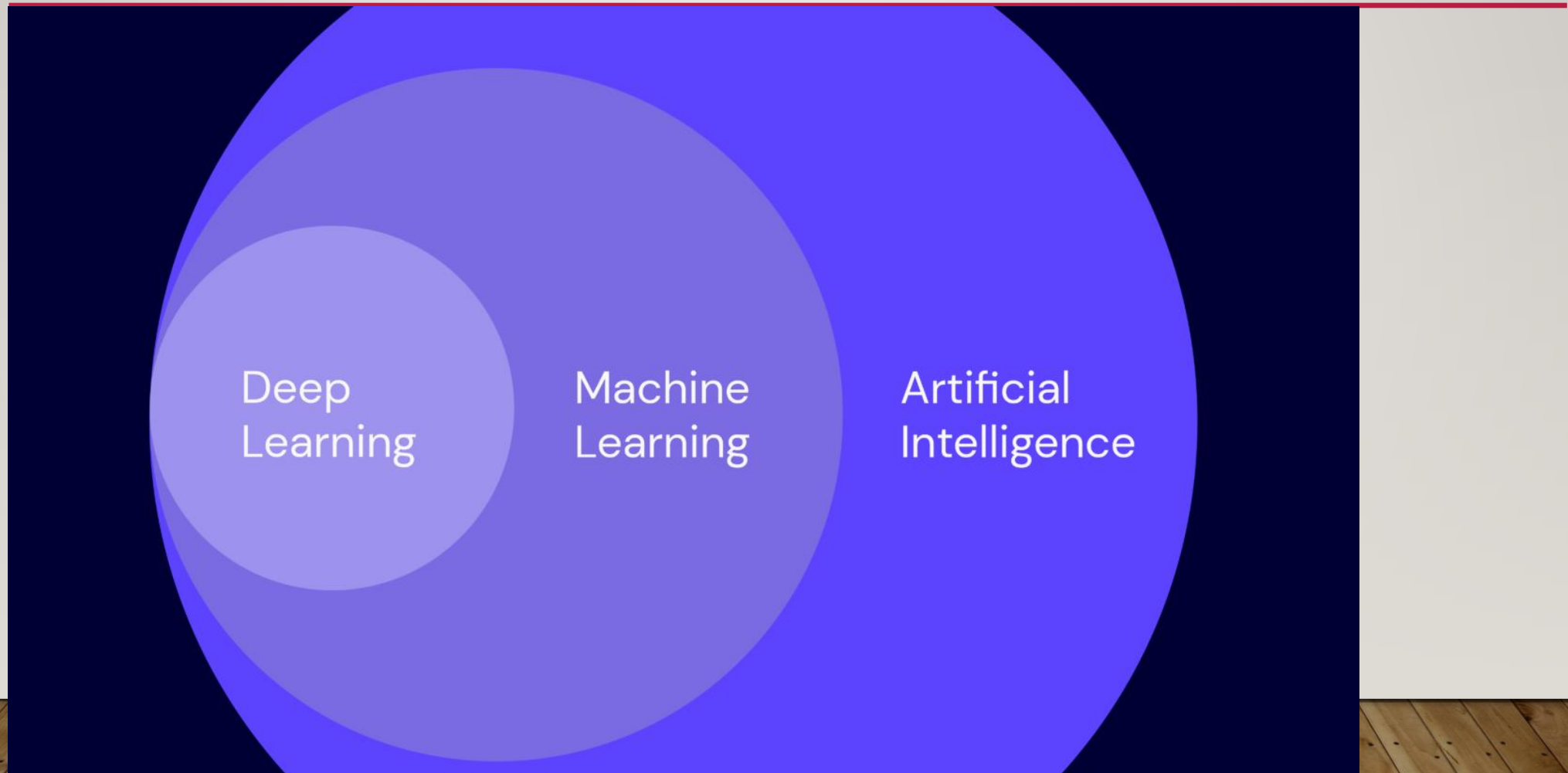
- **Intelligent agents**, which can **perceive their environment** and **take actions** to **achieve goals**, became a focus of AI research.
- **Virtual personal assistants** like Siri and Alexa emerged.
- AI agents became integrated into various **consumer applications**.

10. THE AVAILABILITY OF VERY LARGE DATA SETS (2001–PRESENT):

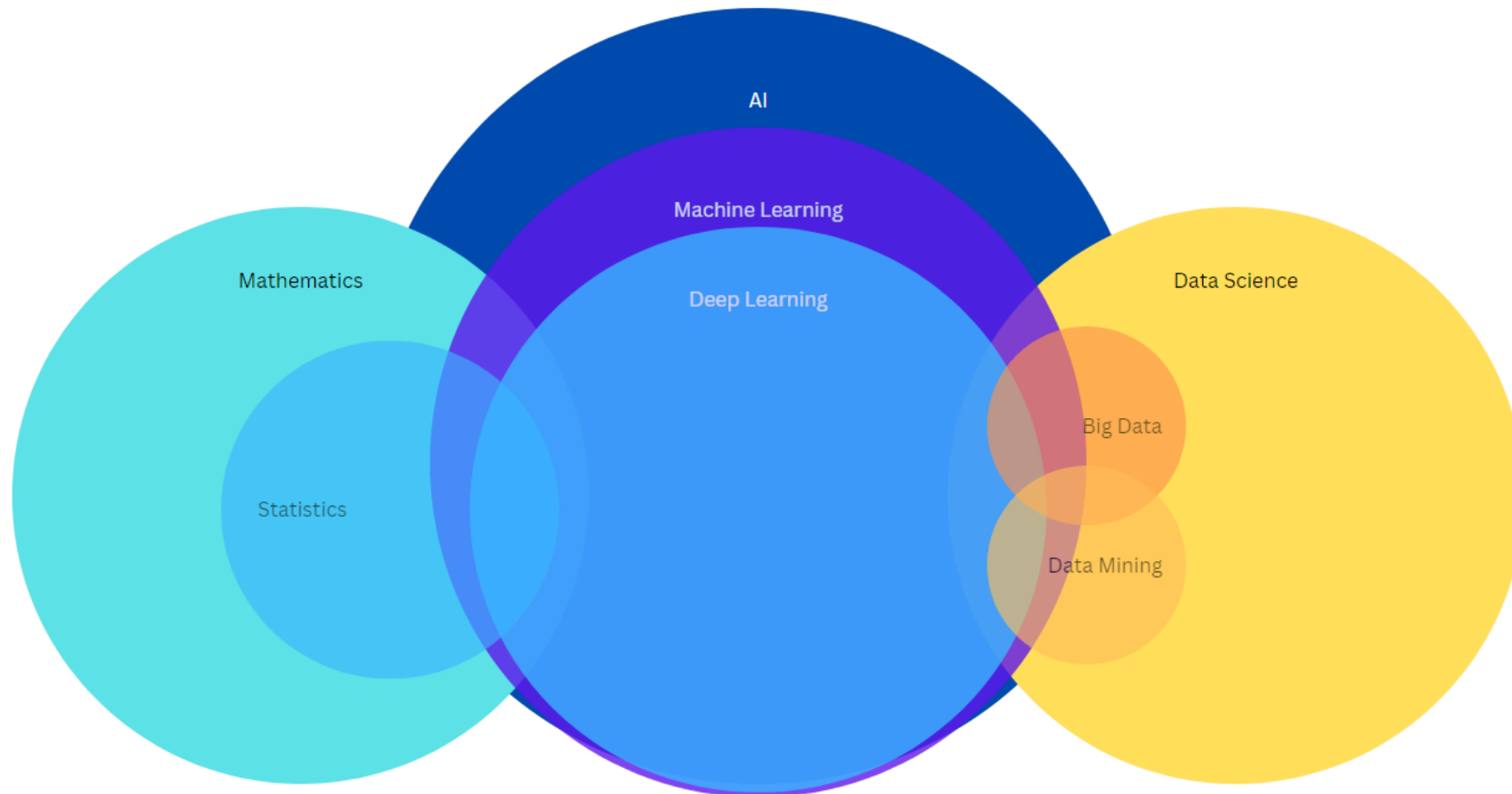
- The availability of **big data** became a **driving force for AI**, enabling more **accurate machine learning models**.
- Technologies like data mining and **deep learning** thrived due to **large datasets**.
- The era of **data-driven AI applications** began, including **recommendation systems** and **predictive analytics**.



AI ML DL



AI, ML, DL, DATA SCIENCE, MATHEMATICS



STATE-OF-THE-ART

- Autonomous planning and scheduling
- Robotic
- Driving
- Image
- Speech
- Video
- Game
- Spam fighting
- Machine Learning

THANK YOU



AI DEFINITION

	Humanly	Rationally
Thinking	1. Thinking Humanly (Lesson Learn)	2.Thinking Rationally (Logic)
Acting	3.Acting Humanly	4.Acting Rationally

THINKING HUMANLY (1/4)



I.THINKING HUMANLY (COGNITIVE SCIENCE)

Human Mind and Brain

We need to **understand** the actual workings of human minds

1. Introspection — catch our own thoughts (**Thinking**)

ตรวจจับความคิดของตัวเอง

Intro + Inspection = วิจัยสนทนา

2. Psychological experiments — observing a person in action (**Behavioral**)

ทำการทดลองทางจิต

สังเกตการแสดงออก และพฤติกรรมของมนุษย์

3. Brain imaging — observing the brain in action (**Brain Work**)

การสร้างภาพของสมอง

สังเกตพฤติกรรมของสมอง

For Example, “General Problem Solver” (Newell and Simon, 1961)

THINKING RATIONALLY (2/4)



2.THINKING RATIONALLY (LOGIC)

arrange a new rule

Aristotle was one of the first to attempt to codify “right thinking”, called **sylogisms**.

- irrefutable reasoning processes

ไม่สามารถพิสูจน์ได้ว่าหลักการผิด

ต้องยอมรับว่าถูก

The syllogism consists of three sentences in which the last sentence is the conclusion

‘All humans must die. We are human. **Therefore, we must die.**’

‘All A are B, and all C are A. **Therefore, all C are B.**’



SYLLOGISMS

1. นิสิต CPE ทุกคน ต้องเรียนวิชา AI

2. Alun Turing เป็นนิสิต CPE

สรุป

ดังนั้น Alun Turing ต้องเรียนวิชา AI

SYLLOGISMS

1. บัณฑิตคณะ ICT@UP ทุกคนต้องผ่านรายวิชาโครงงาน
2. นางเอก ป. ไม่ผ่านรายวิชาโครงงาน

สรุป

ดังนั้น นางเอก ป. ไม่ใช่บัณฑิตคณะ ICT@UP

SYLLOGISMS

What about this sentence?

‘All good programmers write efficient code. Dacha writes efficient code. **Therefore,**
Sarah is a good programmer.’

SYLLOGISMS

What about this sentence?

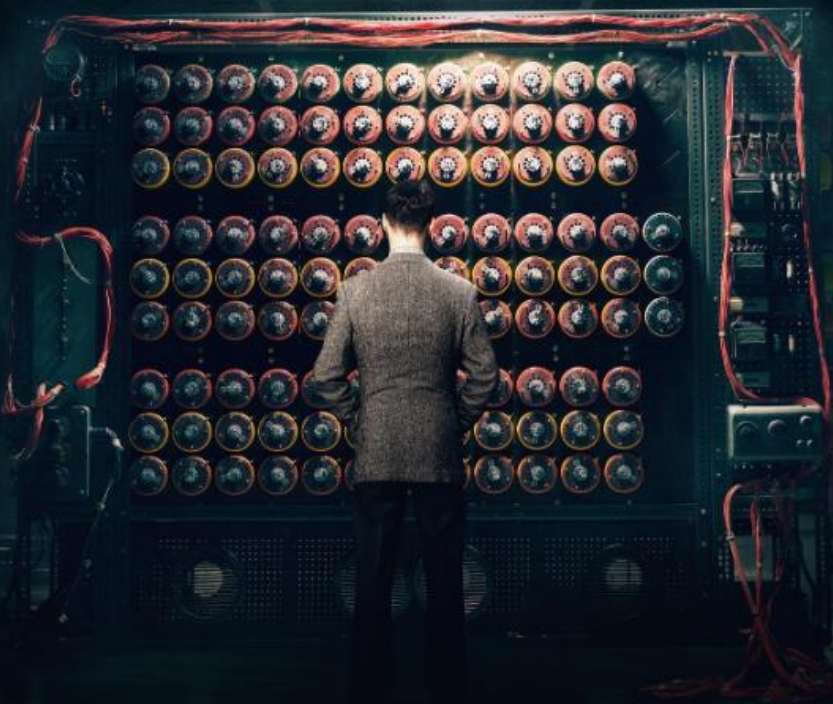
‘All good programmers write efficient code. Dacha writes efficient code. **Therefore,**
Dacha is a good programmer.’

ACTING HUMANLY (3/4)



2014

THE TRUE ENIGMA
WAS THE MAN WHO CRACKED
THE CODE



BENEDICT CUMBERBATCH KEIRA KNIGHTLEY
THE IMITATION GAME

COMING SOON

"IT'S A BRILLIANT FILM!"

-ROGER EBERT

2001



DOUGRAY SCOTT KATE WINSLET JEREMY NORTHAM SAFFRON BURROWS
FROM DIRECTOR **MICHAEL APTED** AND ACADEMY AWARD® WINNING WRITER **TOM STOPPARD**
BASED ON THE BEST SELLING NOVEL BY **ROBERT HARRIS**

E N I G M A

3. ACTING HUMANLY

- The **Turing Test**, proposed by Alan Turing (1950)
- provide a satisfactory operational definition of intelligence
- A computer passes the test if a interrogator ask the question that cannot tell whether the answer come from a person or from a computer
- For now, we note that programming a computer to pass a rigorously applied test provides plenty to work on (LLMAs such as ChatGPT, Bard in 2023)

ให้

มาก

น่าพึงพอใจ

ผู้ซักถาม

สังเกต

Very strong

3. ACTING HUMANLY

The computer would need to possess the following 6 capabilities

- **Natural Language Processing:** enable it to communicate successfully in language
- **Knowledge Representation:** store what it knows or hears
- **Automated Reasoning:** use the stored information to answer questions and to draw new conclusions;
- **Machine Learning:** adapt new circumstances to detect and extrapolate patterns
- **Computer vision** ^{identify} perceive objects (Image, Video)
- **Robotics** ^{เข้าไปยุ่ง} manipulate and move objects

เงื่อนไข

ประมาณค่าภายนอก

เข้าไปยุ่ง

3.ACTING HUMANLY

- **Natural Language Processing:** Google Translate, ChatGPT, Bard
- **Knowledge Representation:** Database and Data Structure
- **Automated Reasoning:** Decision tree, Model Prediction
- **Machine Learning:** Weather Forecast, Traffic Forecast, Background Removal, Face Recognition, Vocal Remover
- **Computer Vision:** background Removal, Face Recognition
- **Robotics:** Humanoid, Robot

ACTING HUMANLY

เข้าใจพื้นฐาน

- **Turing Test**, believing that it is more important to study the underlying principles of intelligence than to duplicate an exemplar

ต้นแบบ

ส่งผ่าน
- Wright brothers and others stopped imitating birds and started using wind tunnels and learning about aerodynamics

เลียนแบบ

ALAN MATHISON TURING

- We can only see a short distance ahead, but we can see plenty there that needs to be done.

ข้างหน้า

Turing, Alan (October 1950), "Computing Machinery and Intelligence" , *Mind*, **LIX** (236): 433–460, doi:10.1093/mind/LIX.236.433

ACTING RATIONALLY (4/4)



ACTING RATIONALLY

Agent: computer agents are expected to operate autonomously, perceive their environment, persist a prolonged time period, adapt to change, and create and pursue goals

Rational Agent: achieve the best outcome no matter when there is uncertainty

ประสบความสำเร็จ

ผลลัพธ์

ไม่แน่นอน

ทำงาน

หยั่งรู้

ล่า

เก็บ

ยึดออก